CSE 327: Software Engineering
Project Idea Proposal Form
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Section: 7

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Project Title: Smart Parking Reservation and Coordination System

Project Description:

Urbanization and the rapid increase in the number of personal vehicles have led to a severe crisis in parking management across busy cities. Drivers often spend a significant amount of time circling commercial areas or hospitals to find a vacant parking spot. This not only wastes valuable time and fuel but also contributes heavily to traffic congestion and carbon emissions. Traditional parking lots still rely on manual ticketing or basic first-come-first-serve models, which completely lack real-time coordination and foresight. To address this real-world problem and align with Sustainable Development Goals (SDG 11: Sustainable Cities and Communities), our team proposes the "Intelligent Parking Reservation and Coordination System (IPRCS)."

IPRCS is not a basic management system; it is an autonomous coordination platform designed to bridge the gap between drivers and parking facility operators using Artificial Intelligence and real-time data. The core objective of this project is to eliminate the uncertainty of finding a parking space. Before leaving for a destination, a user can open our web application, view a live heatmap of available parking slots in their desired area, and reserve a specific spot. The system will autonomously handle the reservation lifecycle, waitlists, and cancellations without requiring any human administrative intervention.

To make the system truly intelligent and robust, we are integrating two distinct AI/ML features that will operate autonomously in the background. First, the system will use a Computer Vision model to monitor live CCTV feeds of the parking lot to accurately count available slots and detect anomalies, ensuring that the digital reservation status matches the physical reality of the lot. Second, the system will utilize statistical machine

learning algorithms to identify suspicious parking behaviors, such as abandoned vehicles that overstay their booked duration by a massive margin.

Ultimately, this project focuses heavily on software engineering design principles, modular architecture, and autonomous decision-making. By implementing continuous integration, automated testing, and agile methodologies, we aim to deliver a scalable product that completely automates parking logistics, optimizes space utilization, and provides a seamless, stress-free experience for drivers.

Technology Stacks:

• Frontend: React.js, Tailwind CSS, TypeScript

• Backend: Python (FastAPI) - Chosen specifically to seamlessly integrate Machine Learning and Computer Vision models.

• Database & Authentication: PostgreSQL, Supabase Auth

• AI/ML & Computer Vision: OpenCV, YOLO (You Only Look Once) model for object detection, Scikit-learn for anomaly detection.

• Version Control & CI/CD: Git, GitHub Actions (for automated linting and testing)

• Deployment: Vercel (Frontend), Render (Backend API), Supabase (Database)

Unique Features
1. Live Free-Slot Counting from Camera Feed (AI/ML Feature) Unlike traditional systems that rely on expensive IoT hardware sensors placed under every single parking space, our system will use a single camera feed to monitor the lot. We will integrate a pre-trained YOLO (You Only Look Once) object detection model via our Python backend. The AI agent will continuously scan the image, count the number of parked cars, and compare them with the predefined slot coordinates. If a car leaves, the AI instantly detects the empty space and updates the real-time database, making the slot available for reservation on the user's app immediately.

2. Abandoned Vehicle Detection System (AI/ML Feature)

In public parking spaces, vehicles are occasionally abandoned or left parked far beyond their reserved time, creating blockages. To solve this, our system will utilize a machine learning anomaly detection algorithm. By calculating the duration z-scores (Standard Score) for each slot type based on historical parking data, the AI agent will identify statistical outliers. If a vehicle's parked duration heavily deviates from the normal distribution and exceeds its booking limit, the system will automatically flag it as an "Abandoned Vehicle" and alert the security admin for towing, ensuring smooth coordination.

3. Automated Grace Period and Dynamic Auto-Release Workflow

To maximize parking lot efficiency and prevent users from holding reservations indefinitely without showing up, we will implement a dynamic auto-release workflow. When a user reserves a slot, the system grants them a specific "Grace Period" (e.g., 15 minutes) to arrive and check in. If the user fails to check in within this time frame, an autonomous background worker will immediately cancel the reservation, apply a penalty policy if applicable, and notify the next user on the waiting list. This ensures high availability and fair coordination among all drivers.

4. Integrated EV (Electric Vehicle) Charging Slot Reservation

As the adoption of Electric Vehicles rises, standard parking spaces are no longer sufficient. Our system will feature dedicated EV charging slot coordination. These specific slots will have their own time-based pricing models based on charging duration rather than just parking duration. Users can filter for EV slots, reserve a charger, and the system will coordinate the turnover time, ensuring that fully charged vehicles do not block the charging station for other incoming EV drivers.